SOLUTIONS
&
SERIAL DILUTIONS

Objectives:

-

1. Define solute, solvent and solution.

2. List and define the types of solution, colloid and crystalloid.

3. Differentiate between emulsion and suspension and give their biomedical / pharmaceutical application.

5. Various important terms used and give their importance in relation to plasma in various clinical situations (electrolytes).

8. Define standard and stock solution.

9. Define serial dilution.

Solutions

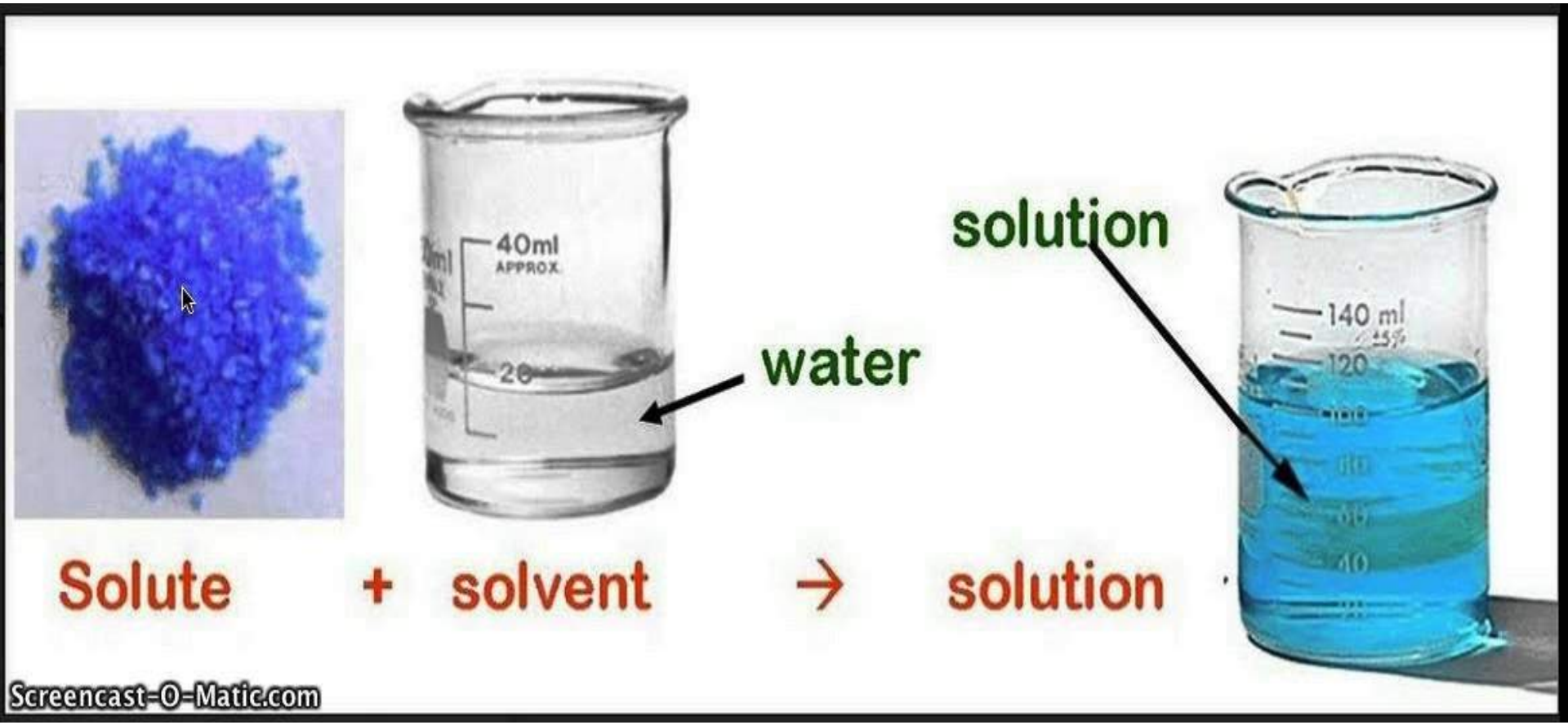
Homogenous mixture of 2 or more substances



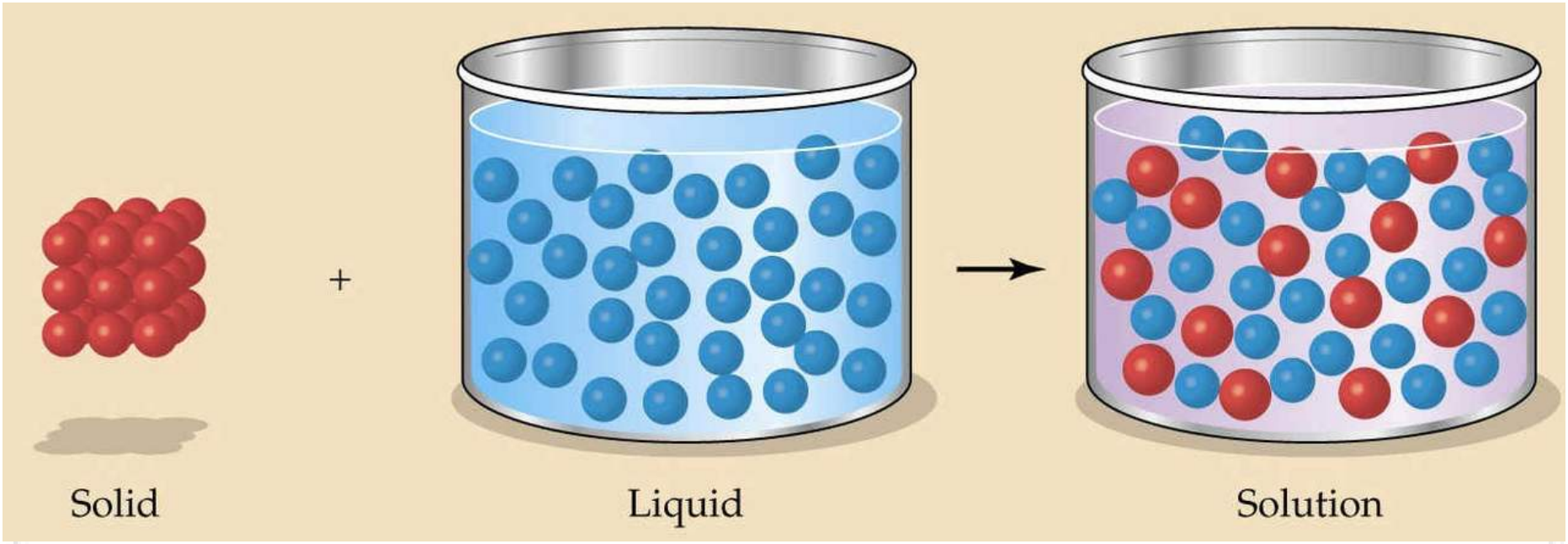
Solute – The substance that dissolves to form a solution

Solvent – The substance in which a solute dissolves

Solution – A mixture of one or more solutes dissolved in a solvent



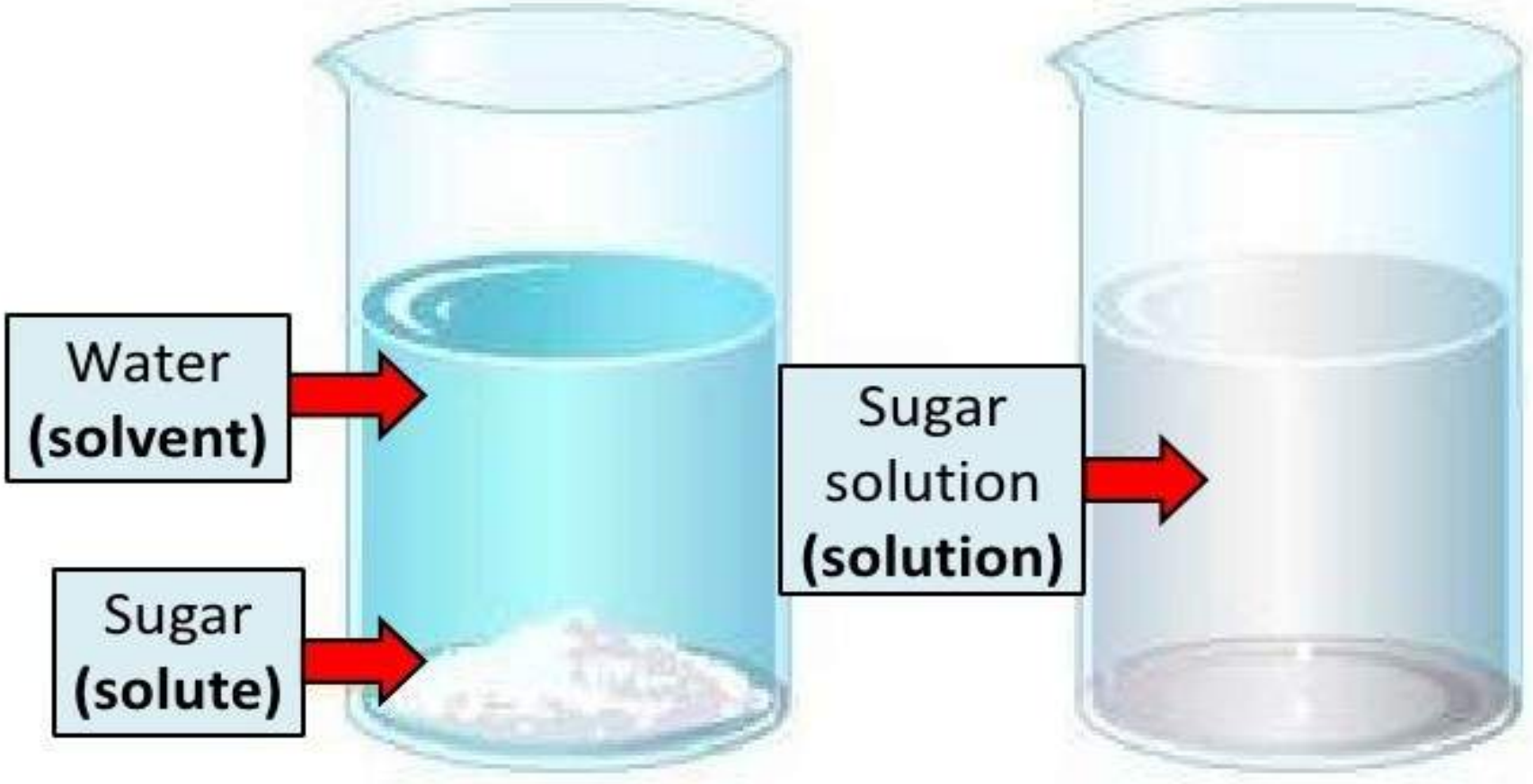
When the solvent is water the solution is said to be **aqueous**


+
Solid
Liquid
Solution


Water
Sugar
Sugar solution
Dissolving of Sugar in Water

SOLUTIONS

A solution is a mixture formed when one or more solutes dissolve in a solvent.



Solution vs. Heterogeneous Mixture



Salt water

solution
homogeneous
stable



Chocolate milk

h. mixture
heterogeneous
will settle

SOLUTION AND SOLUBILITY



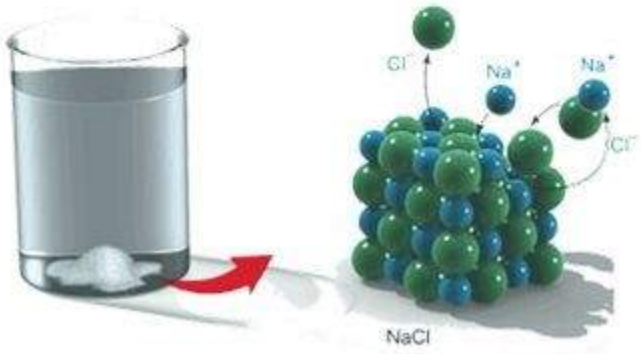
SOLUTE — Substances that dissolves

SOLVENT — Substances that dissolve the solute

Solutions

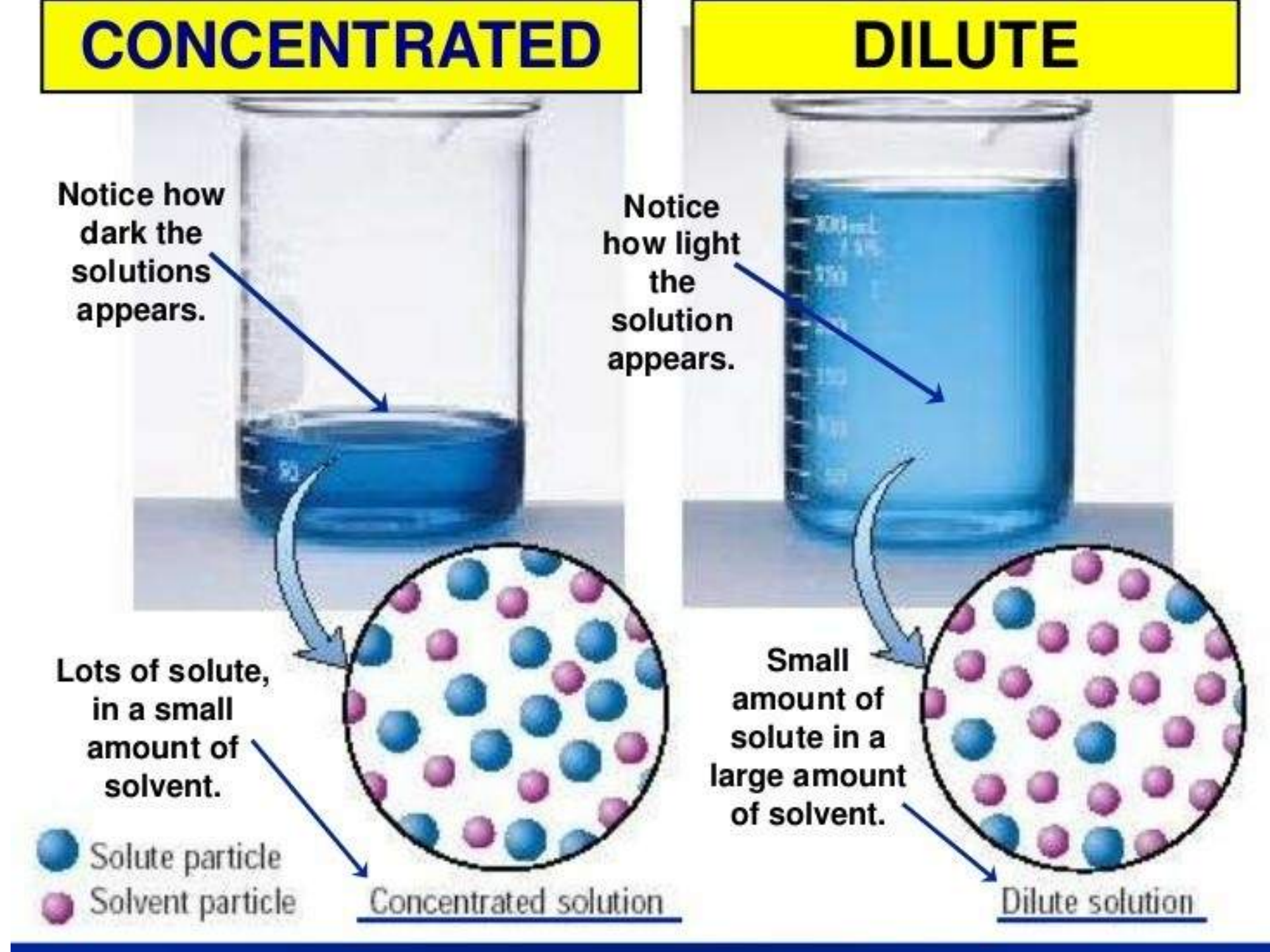
Characteristics of a Solution:

- *Dissolving*
- *Solute*
- *Solvent (Water,…)*
- *Solubility (soluble/insoluble)*





TYPES OF SOLUTIONS

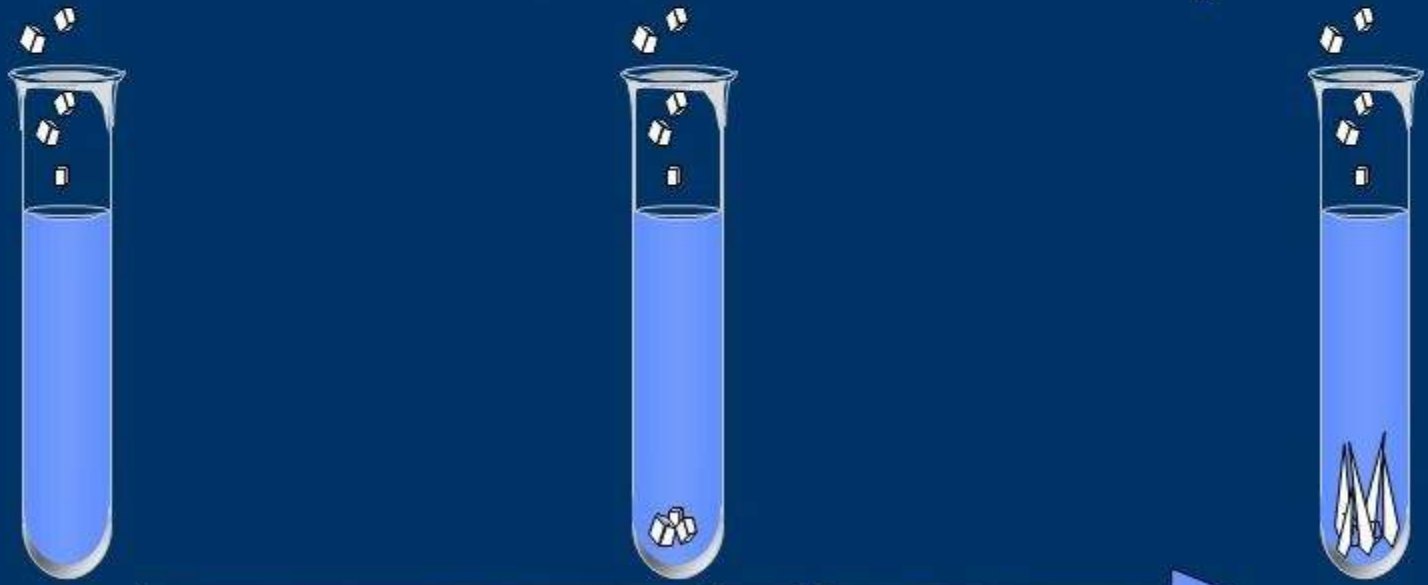

CONCENTRATED
DILUTE
Notice how dark the solutions appears.
Notice how light the solution appears.
Lots of solute, in a small amount of solvent.
Small amount of solute in a large amount of solvent.
Solute particle
Solvent particle
Concentrated solution
Dilute solution

B. Types of Solutions

UNSATURATED SOLUTION
more solute dissolves

SATURATED SOLUTION
no more solute dissolves

SUPERSATURATED SOLUTION
becomes unstable, crystals form



concentration

Three types of solutions

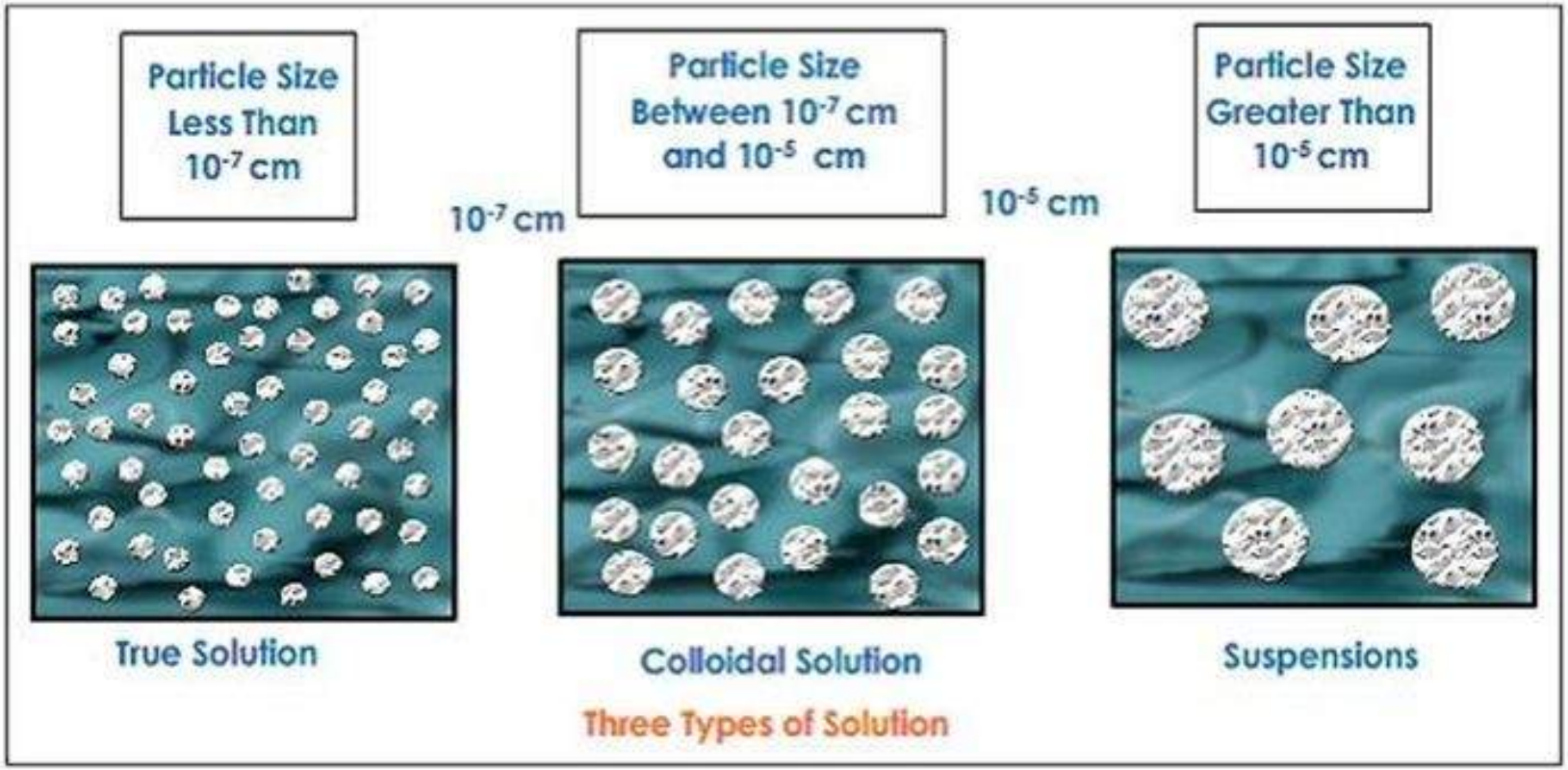
SATURATED SOLUTION: Is a solution that has the maximum amount of solute dissolved in a solvent at a given temperature.

UNSATURATED SOLUTION: Is a solution that can dissolve more solute at a given temperature.

SUPER SATURATED SOLUTION: Is a solution that has more solute than a saturated solution at a given temperature.

$\Rightarrow$ This occurs when a saturated solution is cooled with no "energy source" to precipitate the solute from solution

THREE TYPES OF MIXTURES



SOLUTIONS **COLLOIDS** **SUSPENSIONS**

The **particle size** determines the type of mixture. Solutions and Colloids are **homogeneous**; suspensions are **heterogeneous**.

Different Types of Solutions

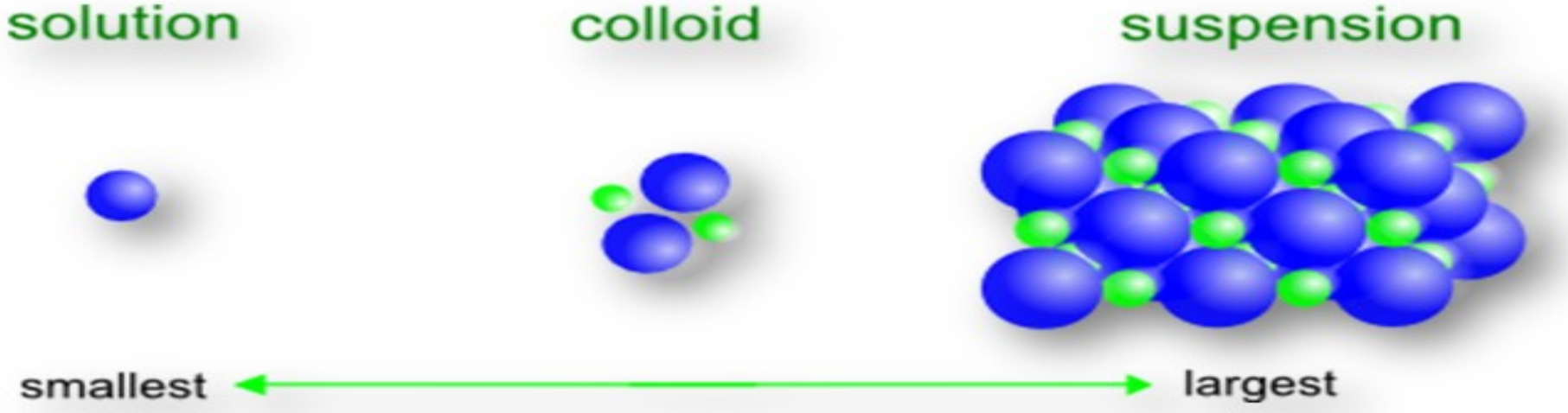
Type of Solution	*Solvent*	*Examples*
Gaseous Solutions	*Gas*	
• Solid in gas		Sublimation of solid in gas ; camphor vapours in N_2 gas
• Liquid in gas		Water vapours in air (mist)
• Gas in gas		Mixture of gases, air
Liquid Solutions	*Liquid*	
• Solid in liquid		Salt in water, sucrose in water
• Liquid in liquid		Alcohol in water, mixture of miscible liquids
• Gas in liquid		Aerated drinks, CO_2 in water
Solid Solutions	*Solid*	
• Solid in solid		Alloys ; homogeneous mixture of two or more metals
• Liquid in solid		Mercury in zinc, mercury in gold
• Gas in solid		Adsorption of gases on metals, solution of hydrogen in palladium

Property	True Solution	Colloidal Solutions	Suspension
Size of the particles	< 1nm	1– 1000nm	>1000nm
Nature	Homogeneous	Homo -to - Heterogeneous	Heterogeneous
Filterability(Diffusion through parchment paper)	Particles of true Solution diffuse rapidly through filter paper as well as parchment paper.	Colloidal particles pass through filter paper but not through parchment paper.	Suspension particles do not pass through filter paper and parchment paper.
Visibility	Particles of True Solution are not visible to naked eye.	Colloidal particles are not seen to naked eye but can be studied through ultra microscope.	Suspension particles are big enough to be seen by naked eye.
Appearance	Transparent	Translucent	Opaque

Crystalloid:

A substance whose particles are smaller than those of a **COLLOID**, form a true solution, and are therefore capable of passing through a **semipermeable MEMBRANE**.

Colloid:
The physical opposite of a crystalloid is a **colloid**,
which does not dissolve and does not form true solutions.



COLLOID vs CRYSTALLOID



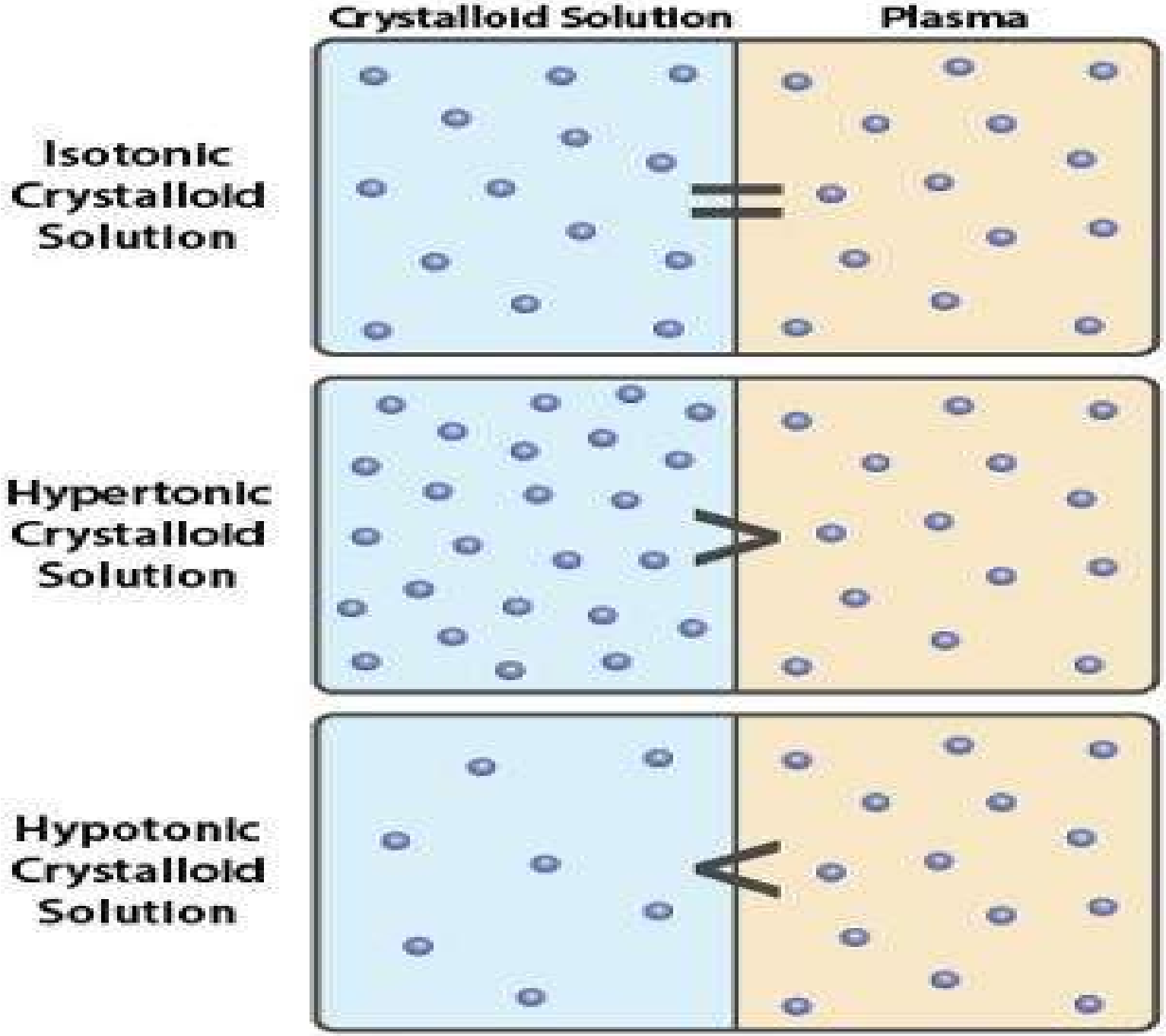
Milk Mixed with Water

Colloid



Salt Dissolved in Water

Crystalloid


Crystalloid Solution
Plasma
Isotonic Crystalloid Solution
Hypertonic Crystalloid Solution
Hypotonic Crystalloid Solution

Crystalloid solutions in common use

Name	Na^+	K^+	Cl^-	HCO_3^-	Ca^{2+}	Glucose
Normal Saline N/S 0.9%	150	0	150	0	0	0
Hartman's	131	5	111	29	2	0
Dextrose Saline D/S / solution 18	30	0	30	0	0	43 g/L
Dextrose 5%	0	0	0	0	0	50 g/L

Colloids in common use

Name	**Pros**	**Cons**
Gelofusin	Made from degraded gelatin (formerly extracted from horses hooves), cheap, no chance of viral transmission, is cleared from the circulation in about 8 hours. No cross match required, on the shelf	Allergic reactions, good for volume but not oxygen carriage
Whole blood	Great for active blood loss,	Hard to come by as the blood bank likes to use this blood to get clotting factors and other goodies. Cross match required, emergency cross match takes 20-40 minutes.
Packed cells	Good to replace haemoglobin,	No clotting factors. Citrate is used as an anticoagulant and binds calcium so the recipient may get citrate toxicity / hypo calcaemia. Risk of viral transmission. Cross match required.
FFP	Replace clotting factors	Expensive, risk of viral transmission. Must be matched to blood type. Once unfrozen has to be given quickly or will go off.

CRYSTALLOIDS OR COLLOIDS?

Crystalloids		Colloids	
Advantages	**Disadvantages**	**Advantages**	**Disadvantages**
• Cheap • Non allergic • No transmission of infection • No interference with coagulation	• Higher volume needed • Relatively short amount of time remaining intravascularly	• Expansion plasma volume far superior • May be salt sparing	• Expensive • Risk of allergy • Coagulopathy • Itch • May exacerbate tissue edema

**The cost of each life saved using crystalloids is $45.13, and the cost of each life saved using colloidal solutions is $1493.60* - http://www.ncbi.nlm.nih.gov/pubmed/2010737

Colloidal System

- The colloidal particles are not visible under microscope but can be observed under an ultramicroscope.
- **Types of Colloidal solutions :**

a) lyophilic (solvent loving) called emulsions, there is an affinity between the particles of the two phases, When water is the dispersion medium it is called hydrophilic



b) lyophobic (solvent hating) called suspensions, When water is the dispersion medium it is called hydrophobic.

An **emulsion** is a mixture of two or more liquids that are normally immiscible (unmixable or unblendable).

Emulsions are part of a more general class of two-phase systems of matter called colloids.

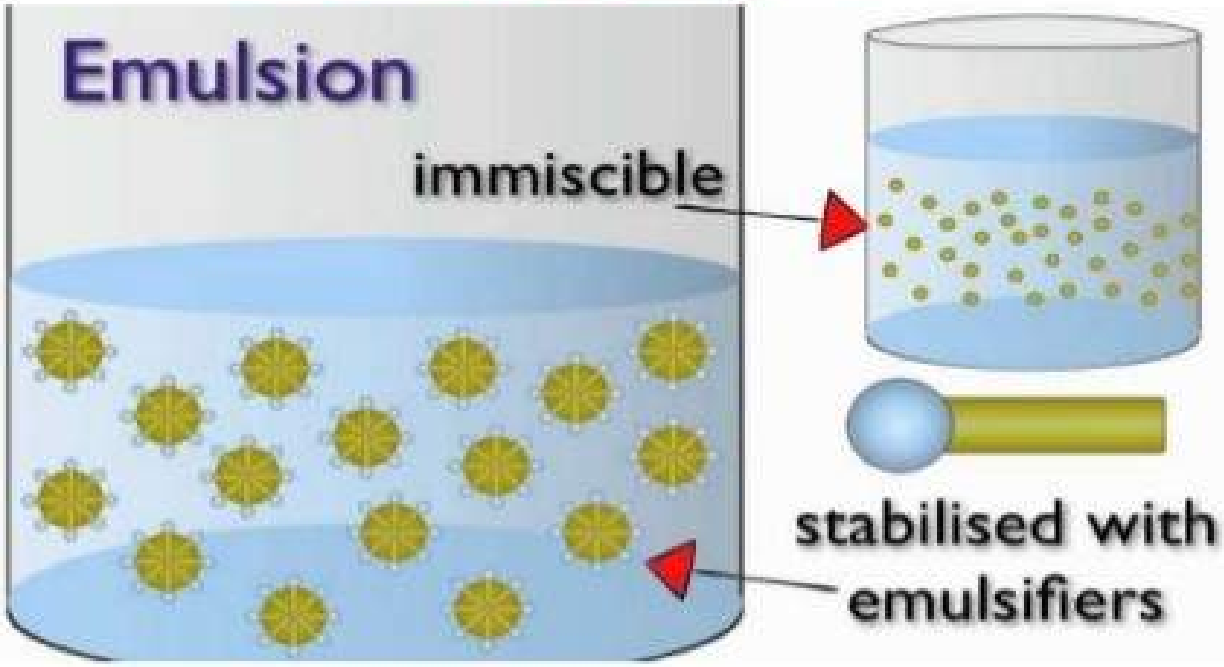
A **suspension** is containing solid particles that are sufficiently large for sedimentation.

Usually they must be larger than one micrometer.

The internal phase (solid) is dispersed throughout the external phase (fluid) through mechanical agitation, with the use of certain excipients or suspending agents.

DEFINITION:

An emulsion is a thermodynamically unstable system consisting of at least two immiscible liquid phases one of which is dispersed as globules in the other liquid phase, stabilized by a third substance called emulsifying agent.

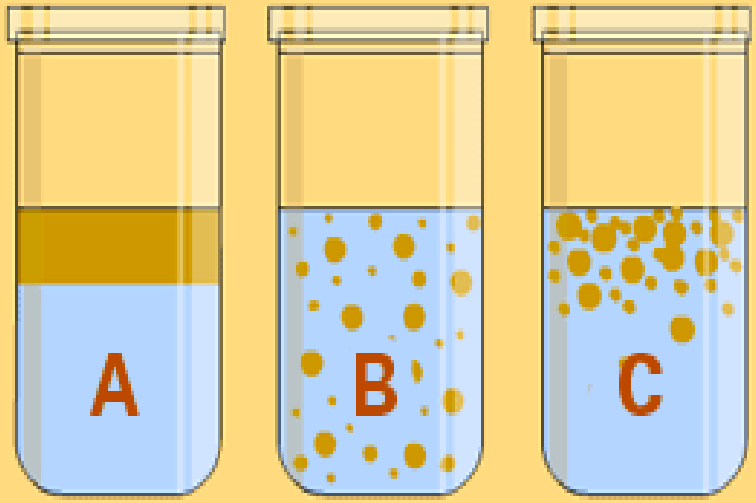


Particle Size

The particle size of the dispersed phase commonly ranges from 0.1 to 100 µm.

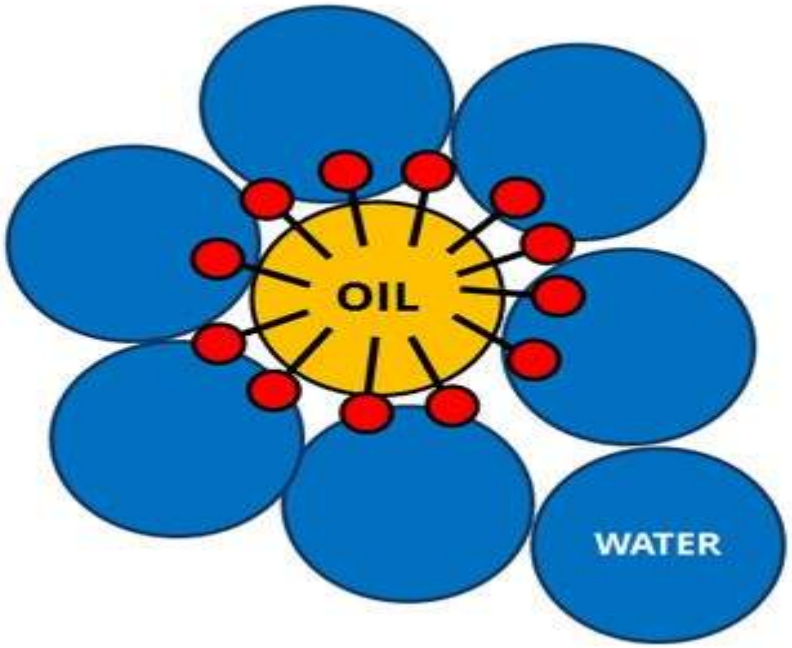
AN **EMULSION** OF OLIVE OIL AND WATER

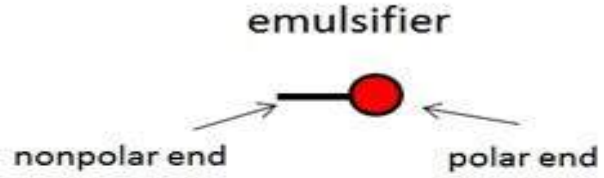
(A) Liquid with higher density usually stays up. The two liquids are immiscible (they don't mix well)

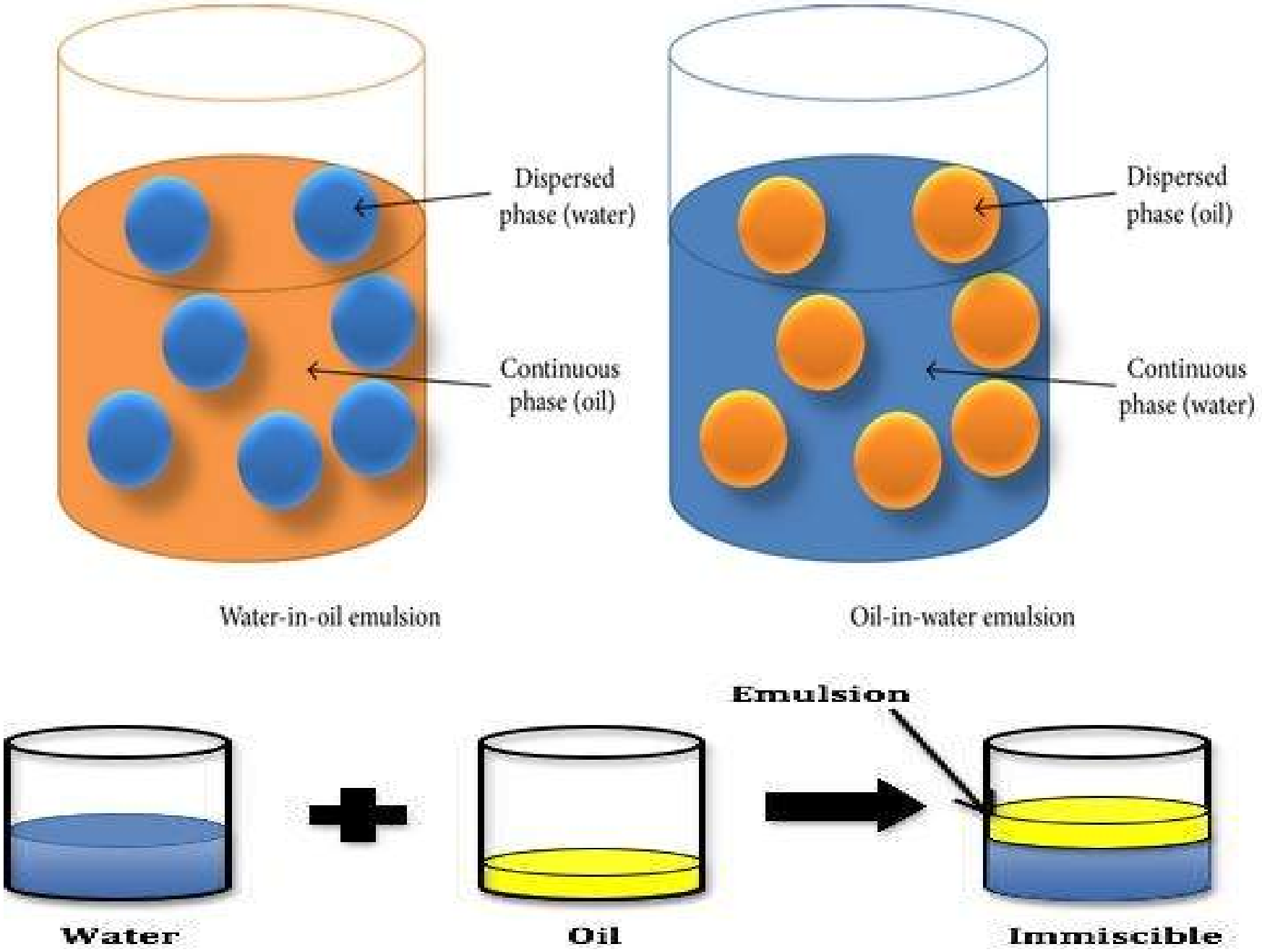


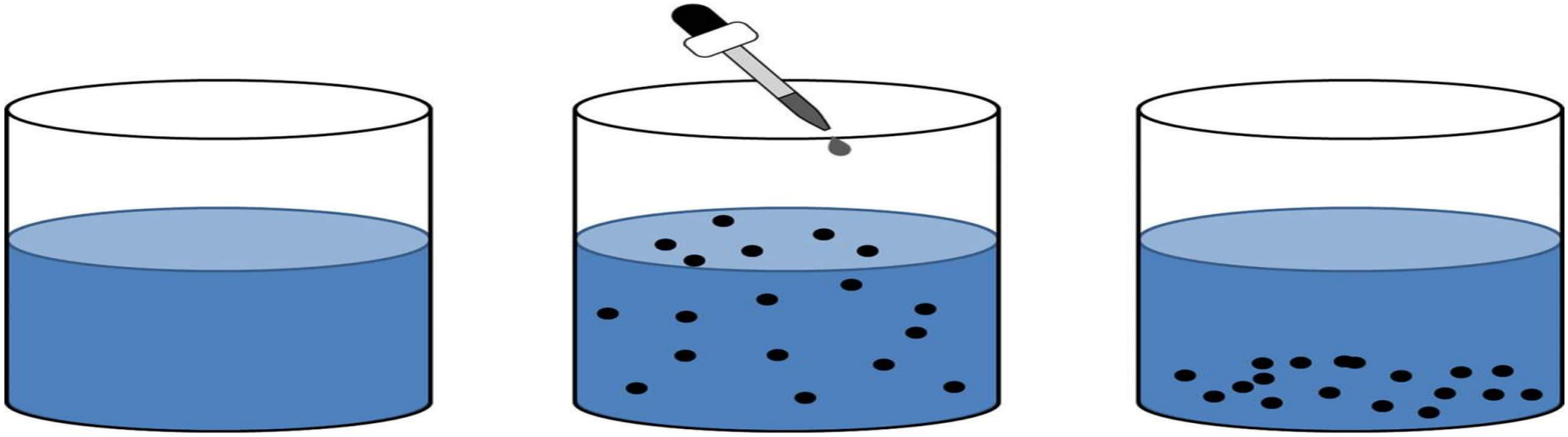
(B) Shake the bottle well and the oil bubble will disperse evenly in the water

(C) After a while, the oil particles begin to come together again at the surface






Dispersed phase (water)
Continuous phase (oil)
Water-in-oil emulsion
Dispersed phase (oil)
Continuous phase (water)
Oil-in-water emulsion
Emulsion
Water
Oil
Immiscible



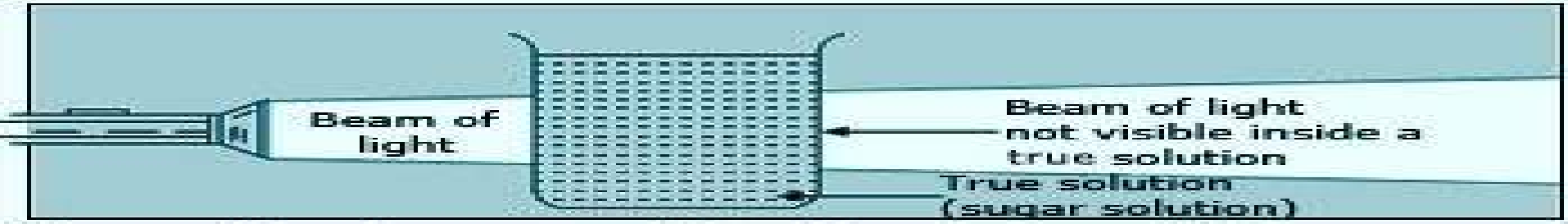
Suspension

Heterogeneous mixtures with dispersed particles as a separate phase.

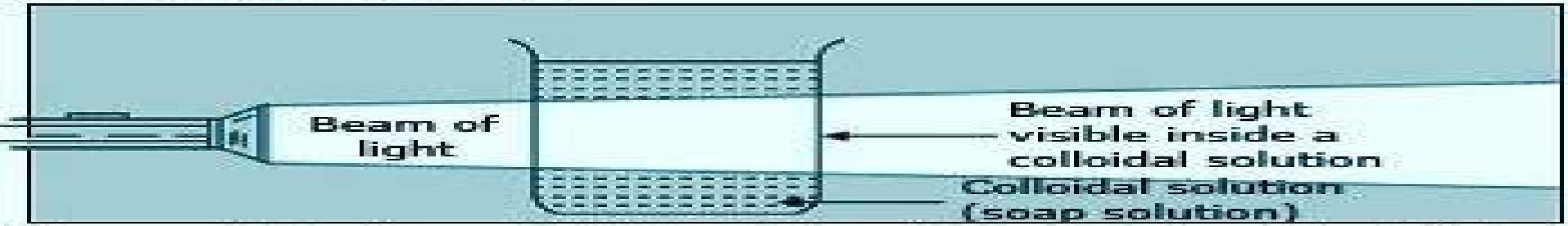


B. Properties of solution, colloids, and suspensions

Solutions	Colloids	Suspensions
Do not separate on standing	**Do not** separate on standing	Particles settle out
Cannot be separated by filtration	Cannot be separated by filtration	**Can** be separated by filtration
Do not scatter light	Scatter light (Tyndall effect)	**May** scatter light, but are not transparent



The particles in a true solution do not scatter light, Due to this we cannot see the path of beam of light inside the true solution (in the beaker)



The particles in a colloidal solution (or colloid) scatter light. Due to this we can see the path of beam of light inside the colloidal solution (or colloid) (in the beaker)

The Tyndall Effect

The scattering of visible light by colloidal particles is called the **Tyndall effect**.





Pharmaceutical Applications of suspensions

Suspensions may be used pharmaceutically for a number of reasons. Some are given bellow;

- Suspension is usually applicable for drug which is insoluble or poorly soluble. E.g. Prednisolone
- suspension To prevent degradation of drug or to improve stability of drug. E.g. Oxytetracycline suspension
- To mask the taste of bitter or unpleasant drug when formulated in solution form. Drugs are formulated in as suspesnion which will be more palatable E.g. Chloramphenicol palmitate suspension
- Suspension of drug can be formulated for topical application e.g. Calamine lotion.

PHARMACEUTICAL APPLICATIONS OF EMULSIONS

1. They can **mask the bitter taste and odor** of drugs, e.g. castor oil, cod-liver oil etc.
2. They can be used to **prolong the release** of the drug thereby providing sustained release action.
3. Essential nutrients like carbohydrates, fats and vitamins can all be emulsified and can be administered to bed ridden patients as sterile **intravenous emulsions**.
4. Emulsions provide **protection to drugs** which are susceptible to oxidation or hydrolysis.
6. Emulsions are used widely to formulate **externally used products** like lotions, creams, liniments etc.

Emulsions encountered in everyday life!



Pesticide



Asphalt



Skin cream



Metal cutting oils



Margarine



Ice cream

Stability of emulsions may be engineered to vary from seconds to years depending on application

Atomic Weight and Number:

- **The atomic number of an atom is simply the number of protons in its nucleus.**

- **The atomic weight of an atom is the mass number of the atom, equal to the total number of protons and neutrons combined.**

Molecular weight:

The sum of the atomic weights of the atoms contained in a molecule. For example, since the atomic weight of hydrogen is 1 and the atomic weight of oxygen is 16, the molecular weight of water (H_2O) is 1 + 1 + 16, or 18.

Equivalent weight:

molecular weight divided by the valency

$$\text{Eq.wt.} = \frac{\text{Mol.wt}}{\text{valency}}$$

Mole:

One mole (mol) of any substance is the **molecular (or atomic) weight** of the substance in **grams**.

One **milli-mole** (mmol) is **0.001 mol** or the **molecular (or atomic) weight** of the substance in **milligrams.**

Osmole:

An **osmole** (osmol) is the amount of a substance that **dissociates** in solution to form **one mole** of osmotically active particles.

A **milliosmole** (mosmol) is **0.001 osmol**, (1/1,000 of an osmole).

A **micro-osmole** (**µOsm**) is 1/1,000,000 of an osmole.

Molarity and molality are both measures of concentration of solutions. One is the **ratio of moles to volume** of the solution and the other is **the ratio of moles to the mass** of the solution.

Molarity, also known as ***molar concentration*,** is the number of *moles of a substance per liter of solution*. Solutions labeled with the molar concentration are denoted with a capital **M.** A 1.0 M solution contains 1 mole of solute per liter of solution.

Molality is the number of ***moles of solute per kilogram of solvent***. It is important the **mass of solvent** is used and not the mass of the solution. Solutions labeled ***with molal concentration*** are denoted with a lower case **m.**

A **1.0 m** solution contains 1 mole of solute per kilogram of **solvent**.

Preparation Of Different types Of solutions

Tonicity:

A property of a solution that depends on the osmotic force exerted across the membrane as influenced by the differing concentrations of solutes in and out of the cell.

It is commonly used when describing the response of cells immersed in an external solution.

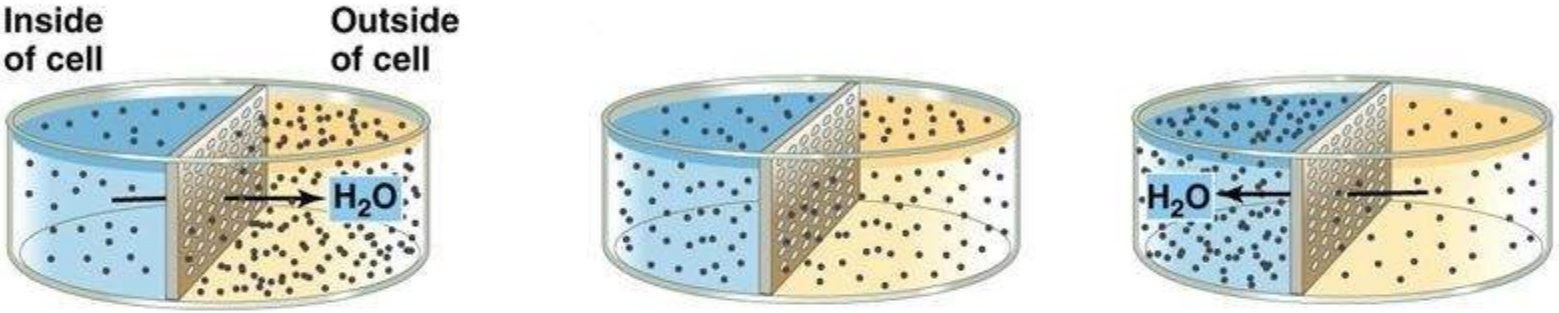
Like osmotic pressure, tonicity is influenced only by solutes that cannot cross the membrane, as only these exert an osmotic pressure.

Osmosis: water moves from a hypotonic solution into a hypertonic solution

(A) **Hypertonic** on the outside (concentrated solutes outside)

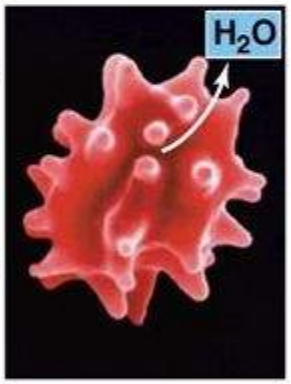
(B) **Isotonic** (equivalent solute concentration)

(C) **Hypotonic** on the outside (dilute solutes outside)



Animal cell (red blood cells)

Cells lose water and shrivel (crenation)



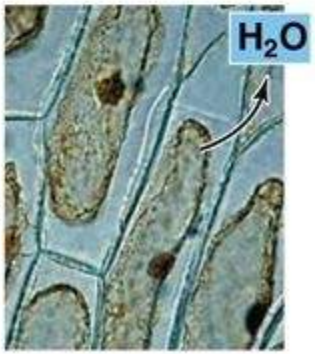




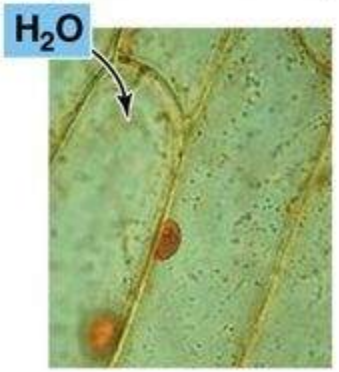
Cells take in water and may burst (hemolysis)

Plant cell (leaf epithelial cells)

Cell body shrinks away from cell wall (wilting)







Cells stiffen and become turgid

- **Concentration** is the measure of how much of a given substance there is mixed with another substance. ***Amount of solute per unit volume of solution.***

- To **concentrate** a solution, one must add more solute (e.g. alcohol), or reduce the amount of solvent (e.g. water).

- To **dilute** a solution, more solvent must be added, or reduce the amount of solute.

1)Percentage Solutions:

- **Can be expressed in three ways:**

a) **W/W% Solution__ Wt. of solute dissolved in 100 gm of solution.**

b) **W/V% Solution__ Wt. of solute dissolved in 100 ml of solution.**

c) **V/V% Solution__ Vol. of solute dissolved in 100 ml of solution.**

2) *Molar Solution:*

- **No. of moles of solute dissolved per liter of solution.**

A commonly used unit for molar concentration is **mol/l**. A solution of concentration 1 mol/l is also denoted as *1 molar* (1 M).

E.g. The molecular weight of sodium chloride (NaCl) is **58.44**, so one gram molecular weight (= 1 mole) is **58.44g**.So to make 1M solution of NaCl dissolve **58.44g** of NaCl in a final volume of 1 liter of solution.

To make a **0.1M NaCl** solution, dissolve 5.844g of NaCl in 1 liter of solution; OR 0.5844g of NaCl in 100ml of solution, OR make a 1:10 dilution of a 1M sample.

3) MOLAL SOLUTION:

A molal solution is a solution that contains No. of moles of solute dissolved in 1000 gm or a kilogram of solvent. The final vol. of solution will be more than 1 liter, (b/c 1000 gm. Water occupies 1liter volume)

$$\text{molality (m)} = \frac{\text{moles of solute}}{\text{1000 gm of solvent}}$$

2. Normal Solutions

- **Normality** is defined as the gram Eq.Wt. of the solute per **L** of the solvent.

  1N sol. = 1 EW solute / 1L of sol.

- Conc. Of acids and alkalis are usually expressed in this unit.

- **gram Eq.Wt.** is the M.W divided by the no. of H^+ or OH^- ions released from 1 molecule of the acid or base, respectively in solutions.

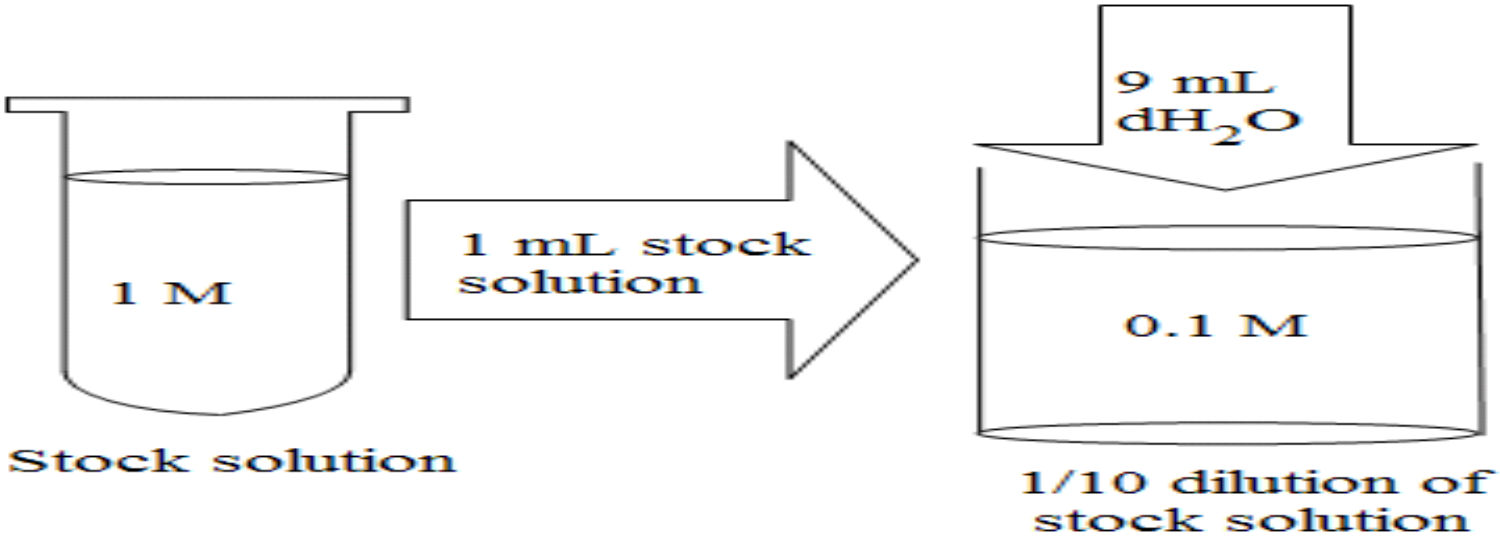
  Eq. Wt. = MW of the substance / replaceable no. of H^+ or OH^-

A **stock solution** is a concentrated solution with which **working** lower concentration solutions (working standards) are prepared.

Stock solutions are used to save preparation time, conserve materials, reduce storage space, and improve the accuracy.

Standard solution:

Standard solution is a solution of known concentration .

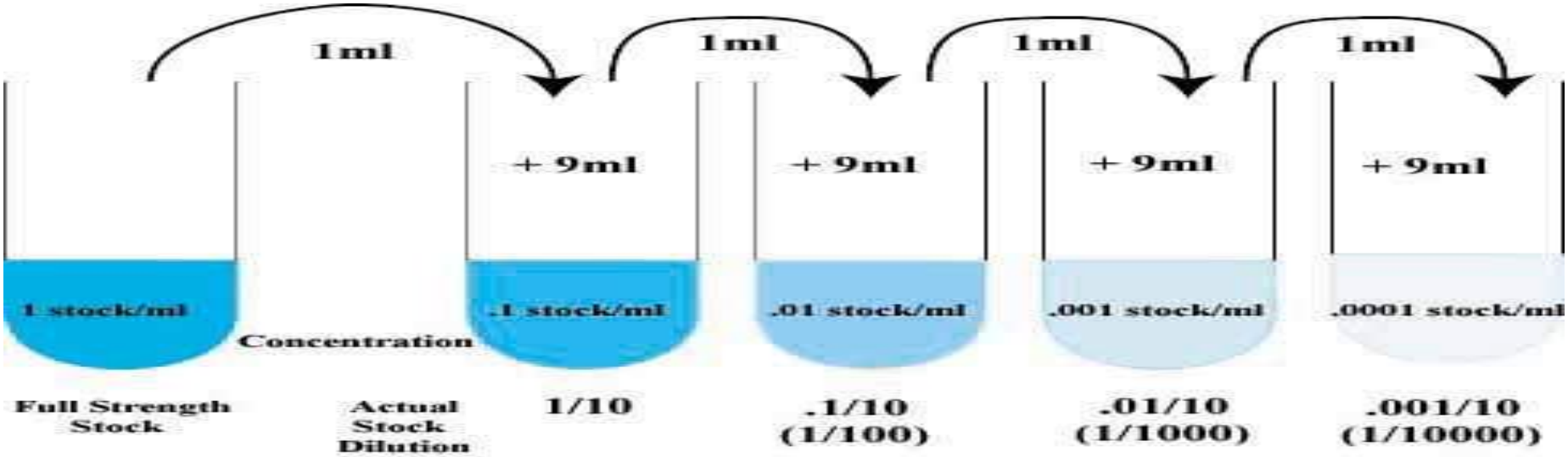


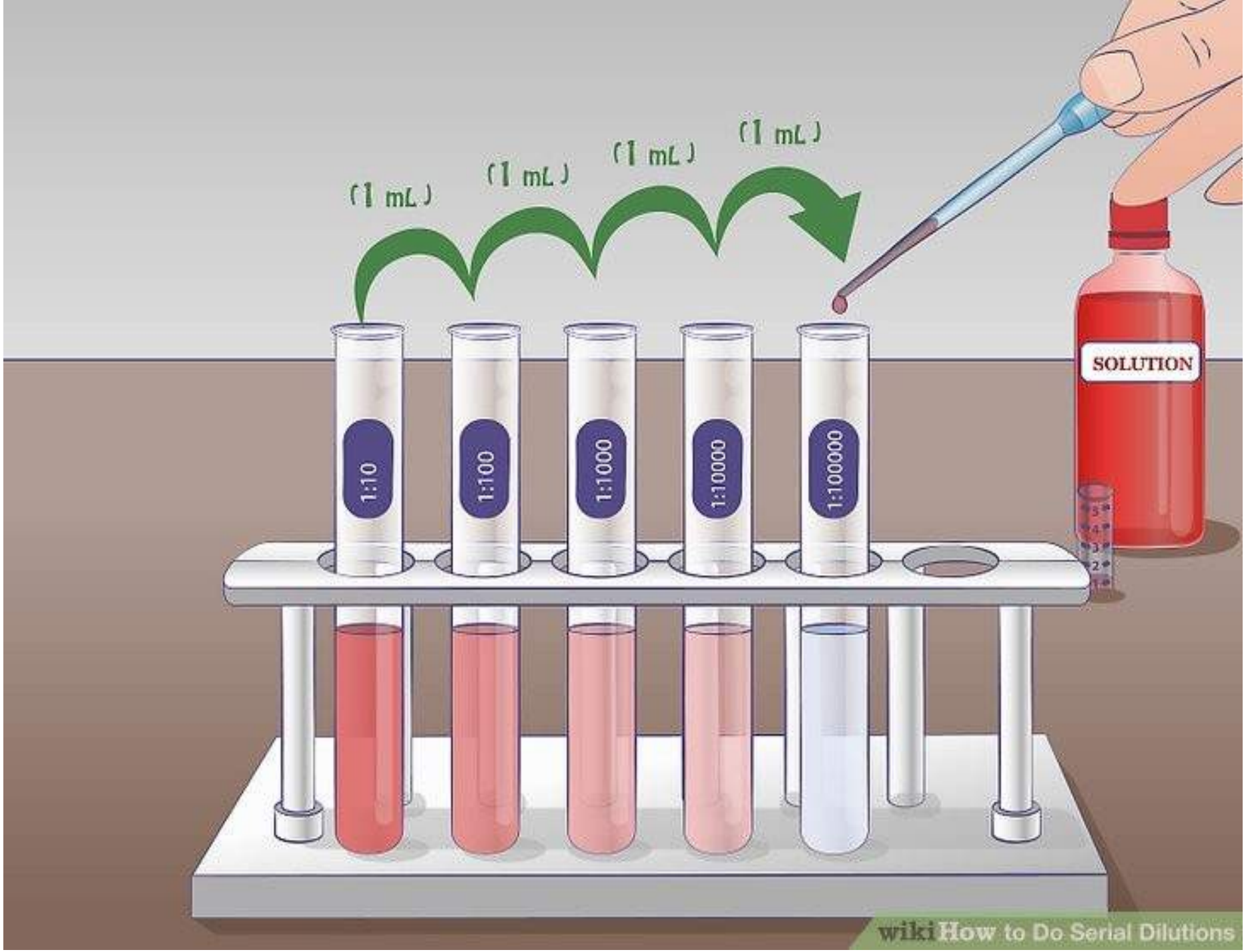
A **serial dilution** is the **stepwise** dilution of a substance in solution.

Usually the dilution factor at each step is constant.

A ten-fold serial dilution could be 1 M, 0.1 M, 0.01 M, 0.001 M, 0.0001M…….

Serial dilutions are used to accurately create highly diluted solutions as well as solutions for experiments resulting in concentration curves.

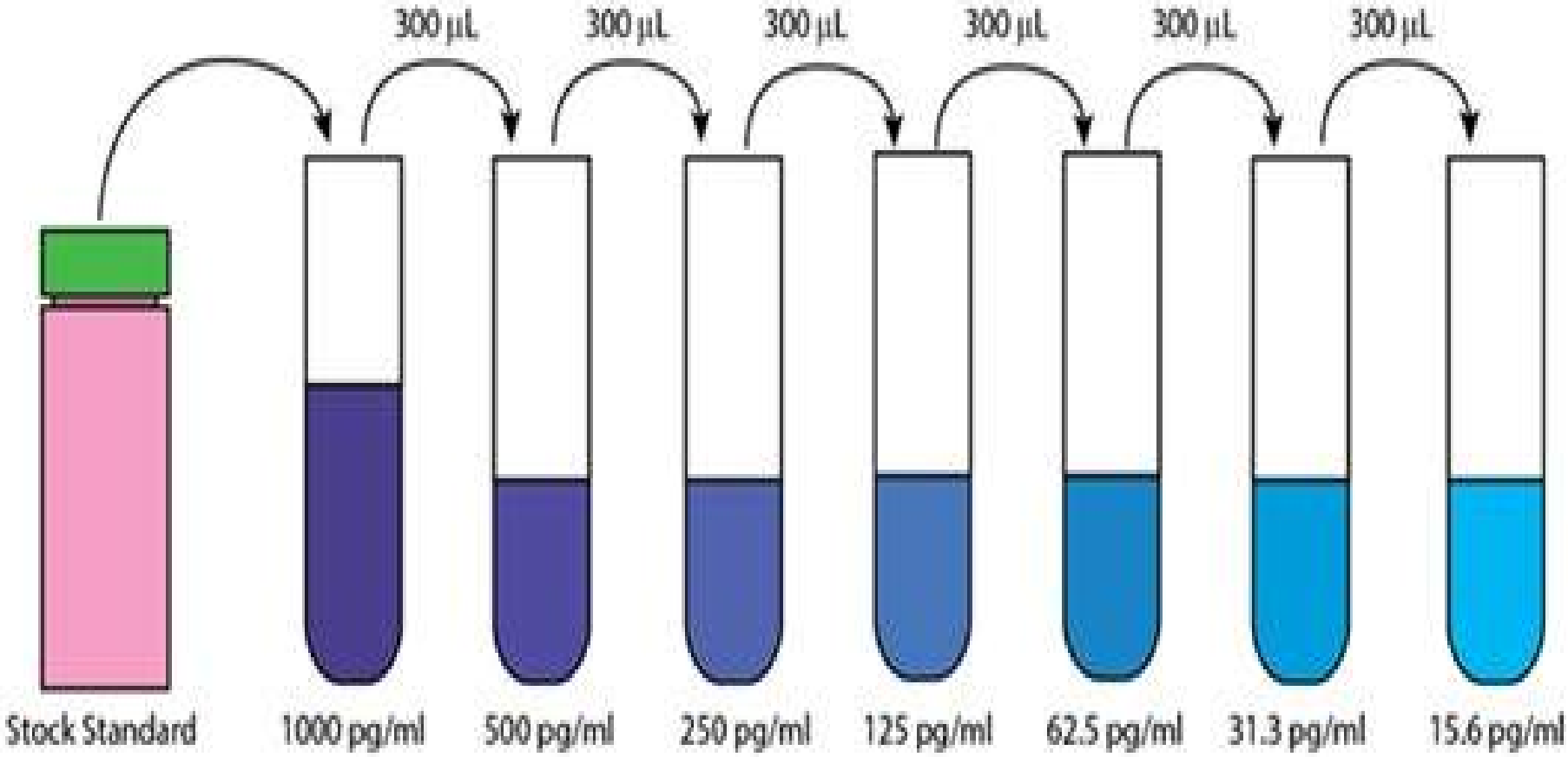




wikiHow to Do Serial Dilutions


1:10
1:100
1:1000
1:10000
1:100000
SOLUTION
Dt = D1 x D2 x D3 x ... x Dnt
Dt = 10 x 10 x 10 x 10 = 10,000
Cfinal = Cinitial/D
Cfinal = 100,000/1,000
Cfinal = 100 cells per mL.

Why do we need serial dilutions?



What is Saline Solution?

1. In medicine, saline (also saline solution) is a solution of sodium chloride (NaCl, table salt) in water.
2. Commonly used for a solution of 0.90% w/v of NaCl, 308 mOsm/L or 9.0 g per litre.
3. This solution is referred to as physiological saline or isotonic saline (because it closely approximates isotonic, that is, physiologically normal, solution) i.e., good fluid balance with minimal hypotonicity or hypertonicity.
4. It is used to flush wounds and skin abrasions, as eye drops, for intravenous infusion, rinsing contact lenses, nasal irrigation, and a variety of other purposes.

To prepare of 1000 (1 liter) of normal saline.

0.9% NaCl solution

OR 0.9 gm in 100 ml

100 ml normal saline contains 0.9 gm

1 ml normal saline container (0.9/100) gm

1000 ml normal saline container (0.9*1000)/100 gm= 9 gm

Apparatus needed :

1. Chemical balance with weight box.
2. Volumetric flask / 1000 ml
3. Stirrer
4. Reagent bottle 1000 ml
5. Funnel
6. White paper

Procedure for 0.9% Saline (in 1000 ml):

1. Weight 9 gm of Sodium chloride (NaCl) with the help of balance.
2. Take 500 ml of distilled water in a beaker/volumetric flask and added 9 gm NaCl.
3. Mixed it properly then distilled water is added up to 1000 ml.
4. Then again mixed it with the help of stirrer. Transferred it to the reagent bottle and labeled.
5. Now 1000 ml of normal saline is prepared.

Prepare Dextrose (5%) Solution:

Dextrose 5% solution = 5 gm in 100 ml

For 1 Litre OR 1000ml

50 g dextrose in 1000ml

DEXTROSE 5%

- *Isotonic solution - No electrolytes*
- *50 g/L of glucose*
- *Provide modest calories (1L - 200kcal)*
- *Rapidly metabolized & distribute evenly throughout the all compartments*




THE END